



Power management

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Category: Power management

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Power management  is a feature that turns off the power or switches system components to a low-power state when inactive.

In Arch Linux, power management consists of two main parts:

1. Configuration of the Linux kernel, which interacts with the hardware.
 - [Kernel parameters](#)
 - [Kernel modules](#)
 - [udev](#) rules
2. Configuration of userspace tools, which interact with the kernel and react to its events. Many userspace tools also allow to modify kernel configuration in a "user-friendly" way. See [#Userspace tools](#) for the options.

1 Userspace tools

Using these tools can replace setting a lot of settings by hand. Only run **one** of these tools to avoid possible conflicts as they all work more or less similarly. Have a look at the [power management category](#) to get an overview on what power management options exist in Arch Linux.

These are the more popular scripts and tools designed to help power saving:

1.1 Console

- [acpid](#) — A daemon for delivering ACPI power management events with netlink support.
<https://sourceforge.net/projects/acpid2/>  || [acpid](#)
- [Laptop Mode Tools](#) — Utility to configure laptop power saving settings, considered by many to be the de facto utility for power saving though may take a bit of configuration.
<https://github.com/rickysarraf/laptop-mode-tools>  || [laptop-mode-tools](#)^{AUR}
- [libsmbios](#) — Library and tools for interacting with Dell SMBIOS tables.

<https://github.com/dell/libsmbios> || **libsmbios**

- **powertop** — A tool to diagnose issues with power consumption and power management to help set power saving settings.

<https://github.com/fenrus75/powertop> || **powertop**

- **powerstat** — A tool that measures power consumption using the ACPI or Intel RAPL interface.

<https://github.com/ColinIanKing/powerstat> || **powerstat**^{AUR}

- **systemd** — A system and service manager.

<https://systemd.io/> || **systemd**

- **TLP** — Advanced power management for Linux.

<https://linrunner.de/tlp> || **t1p**

- **Tuned** — Daemon that performs monitoring and adaptive configuration of devices in the system

<https://tuned-project.org> || **tuned**^{AUR}

- **upower** — Abstraction for enumerating power devices, listening to device events and querying history and statistics.

<https://upower.freedesktop.org> || **upower**

1.2 Graphical

- **batsignal** — Lightweight battery monitor that uses libnotify to warn of low battery levels.

<https://github.com/electrickite/batsignal> || **batsignal**^{AUR}

- **cbatticon** — Lightweight and fast battery icon that sits in your system tray.

<https://github.com/valr/cbatticon> || **cbatticon**

- **GNOME Power Statistics** — System power information and statistics for GNOME.

<https://gitlab.gnome.org/GNOME/gnome-power-manager> || **gnome-power-manager**

- **KDE Power Devil** — Power management module for Plasma.

<https://invent.kde.org/plasma/powerdevil> || **powerdevil**

- **LXQt Power Management** — Power management module for LXQt.

<https://github.com/lxqt/lxqt-powermanagement> || **lxqt-powermanagement**

- **MATE Power Management** — Power management tool for MATE.

<https://github.com/mate-desktop/mate-power-manager> || **mate-power-manager**

- **MATE Power Statistics** — System power information and statistics for MATE.

<https://github.com/mate-desktop/mate-power-manager> || **mate-power-manager**

- **poweralertd** — Daemon for delivering UPower notifications.

<https://git.sr.ht/~kennylevinsen/poweralertd> || **poweralertd**^{AUR}

- **powerkit** — Desktop independent power manager.

<https://github.com/rodlie/powerkit> || **powerkit**^{AUR}

- **Xfce Power Manager** — Power manager for Xfce.

<https://docs.xfce.org/xfce/xfce4-power-manager/start> || **xfce4-power-manager**

- **vattery** — Battery monitoring application written in Vala that will display the status of a laptop battery in a system tray.

<https://www.jezra.net/projects/vattery.html> || [vattery](#)^{AUR}

2 ACPI events

`systemd` handles some power-related [ACPI](#) events, whose actions can be configured in `/etc/systemd/logind.conf` or `/etc/systemd/logind.conf.d/*.conf` — see [logind.conf\(5\)](#). On systems with no dedicated power manager, this may replace the `acpid` daemon which is usually used to react to these ACPI events.

The specified action for each event can be one of `ignore`, `poweroff`, `reboot`, `halt`, `suspend`, `hibernate`, `hybrid-sleep`, `suspend-then-hibernate`, `lock` or `kexec`. In case of hibernation and suspension, they must be properly [set up](#). If an event is not configured, `systemd` will use a default action.

Event handler	Description	Default action
<code>HandlePowerKey</code>	Triggered when the power key/button is pressed.	<code>poweroff</code>
<code>HandleSuspendKey</code>	Triggered when the suspend key/button is pressed.	<code>suspend</code>
<code>HandleHibernateKey</code>	Triggered when the hibernate key/button is pressed.	<code>hibernate</code>
<code>HandleLidSwitch</code>	Triggered when the lid is closed, except in the cases below.	<code>suspend</code>
<code>HandleLidSwitchDocked</code>	Triggered when the lid is closed if the system is inserted in a docking station, or more than one display is connected.	<code>ignore</code>
<code>HandleLidSwitchExternalPower</code>	Triggered when the lid is closed if the system is connected to external power.	action set for <code>HandleLidSwitch</code>

To apply changes, [reload](#) `systemd-logind.service`.

Note:

- `systemd` cannot handle AC and Battery ACPI events, so if you use [Laptop Mode Tools](#) or other similar tools `acpid` is still required.
- When performing lid switches in short succession, `logind` will delay the suspend action for up to 90s to detect possible docks. [\[1\]](#) This delay was made configurable with `systemd` v220. [\[2\]](#) You can use e.g. `HoldoffTimeoutSec=30s` in `logind.conf` or its drop-in file.

2.1 Power managers

Some [desktop environments](#) include power managers which [inhibit](#) (temporarily turn off) some or all of the `systemd` ACPI settings. If such a power manager is running, then the actions for ACPI events can be configured in the power manager alone. Changes to `/etc/systemd/logind.conf` or `/etc/systemd/logind.conf.d/*.conf` need be made only if you wish to configure behaviour for a particular event that is not inhibited by the power manager.

Note that if the power manager does not inhibit *systemd* for the appropriate events you can end up with a situation where *systemd* suspends your system and then when the system is woken up the other power manager suspends it again. The power managers of **KDE**, **GNOME**, **Xfce** and **MATE** issue the necessary *inhibited* commands. If the *inhibited* commands are not being issued, such as when using **acpid** or others to handle ACPI events, set the `Handle` options to `ignore`. See also **systemd-inhibit(1)**.

2.2 xss-lock

xss-lock subscribes to the *systemd-events* `suspend`, `hibernate`, `lock-session`, and `unlock-session` with appropriate actions (run locker and wait for user to unlock or kill locker). *xss-lock* also reacts to **DPMS** events and runs or kills the locker in response.

Autostarting the following for example:

```
$ xss-lock -- i3lock -n -i background_image.png &
```

3 Power saving

Note: See **Laptop#Power management** for power management specific to laptops, such as battery monitoring. See also pages specific to your CPU and GPU (e.g., **Ryzen**, **AMDGPU**).

This section is a reference for creating custom scripts and power saving settings such as by udev rules. Make sure that the settings are not managed by some **other utility** to avoid conflicts.

Almost all of the features listed here are worth using whether or not the computer is on AC or battery power. Most have negligible performance impact and are just not enabled by default because of commonly broken hardware/drivers. Reducing power usage means reducing heat, which can even lead to higher performance on a modern Intel or AMD CPU, thanks to **dynamic overclocking**.

3.1 Processors with Intel HWP (Intel Hardware P-state) support



This article or section is a candidate for merging with **CPU frequency scaling.**

Notes: More context in the main article. (Discuss in **Talk:Power management**)

The available energy preferences of a HWP supported processor are `default`, `performance`, `balance_performance`, `balance_power`, `power`.

This can be validated by running

```
$ cat /sys/devices/system/cpu/cpufreq/policy*/energy_performance_available_preferences
```

To conserve more energy, you can configuration by creating the following file:

```
/etc/tmpfiles.d/energy_performance_preference.conf
```

```
w /sys/devices/system/cpu/cpufreq/policy*/energy_performance_preference - - - - balance
_power
```

See the **x86_energy_perf_policy(8)** man page for more details on energy-performance policy in Intel

processors. Also see [systemd-tmpfiles\(8\)](#) and [tmpfiles.d\(5\)](#) man pages for temporary files/directories details.

3.2 Audio

3.2.1 Kernel

By default, audio power saving is turned off by most drivers. It can be enabled by setting the `power_save` parameter; a time (in seconds) to go into idle mode. To idle the audio card after one second, create the following file for Intel soundcards.

```
/etc/modprobe.d/audio_powersave.conf

options snd_hda_intel power_save=1
```

Alternatively, use the following for ac97:

```
options snd_ac97_codec power_save=1
```

Note:

- To retrieve the manufacturer and the corresponding kernel driver which is used for your sound card, run `lspci -k`.
- Toggling the audio card's power state can cause a popping sound or noticeable latency on some broken hardware.

It is also possible to further reduce the audio power requirements by disabling the HDMI audio output, which can be done by [blacklisting](#) the appropriate kernel modules (e.g. `snd_hda_codec_hdmi` in case of Intel hardware).

3.2.2 Pops when starting and stopping playback

By default, audio sources that have become idle for too long are suspended. When playing a sound or using a microphone, playback or recordings may start with a pop sound. See the dedicated pages for the relevant solution:

- [Advanced Linux Sound Architecture/Troubleshooting#Pops when starting and stopping playback](#)
- [PulseAudio/Troubleshooting#Pops when starting and stopping playback](#)
- [PipeWire#Noticeable audio delay or audible pop/crack when starting playback](#)

3.3 Backlight

See [Backlight](#).

3.4 Bluetooth

To disable bluetooth completely, [blacklist](#) the `btusb` and `bluetooth` modules.

To turn off bluetooth only temporarily, use *rfkill*:

```
# rfkill block bluetooth
```

Or with udev rule:

```
/etc/udev/rules.d/50-bluetooth.rules
```

```
# disable bluetooth
SUBSYSTEM=="rfkill", ATTR{type}=="bluetooth", ATTR{state}="0"
```

3.5 Web camera

If you will not use integrated web camera then **blacklist** the `uvcvideo` module.

3.6 Kernel parameters

This section uses configurations in `/etc/sysctl.d/`, which is *"a drop-in directory for kernel sysctl parameters."* See [The New Configuration Files](#) and more specifically [sysctl.d\(5\)](#) for more information.

3.6.1 Disabling NMI watchdog



This article or section needs expansion.

Reason: This or `nowatchdog` as can be seen in [Improving performance#Watchdogs](#) (Discuss in [Talk:Power management](#))

The **NMI** [watchdog](#) is a debugging feature to catch hardware hangs that cause a kernel panic. On some systems it can generate a lot of interrupts, causing a noticeable increase in power usage:

```
/etc/sysctl.d/disable_watchdog.conf
```

```
kernel.nmi_watchdog = 0
```

or add `nmi_watchdog=0` to the **kernel line** to disable it completely from early boot.

3.6.2 Writeback Time

Increasing the virtual memory dirty writeback time helps to aggregate disk I/O together, thus reducing spanned disk writes, and increasing power saving. To set the value to 60 seconds (default is 5 seconds):

```
/etc/sysctl.d/dirty.conf
```

```
vm.dirty_writeback_centisecs = 6000
```

To do the same for journal commits on supported filesystems (e.g. ext4, btrfs...), use `commit=60` as a option in **fstab**.

Note that this value is modified as a side effect of the Laptop Mode setting below. See also [sysctl#Virtual memory](#) for other parameters affecting I/O performance and power saving.

3.6.3 Laptop Mode

See the [kernel documentation](#) on the laptop mode "knob" - "A sensible value for the knob is 5 seconds".

```
/etc/sysctl.d/laptop.conf
```

```
vm.laptop_mode = 5
```

Note: This setting is mainly relevant to spinning-disk drives.

3.7 Network interfaces

Wake-on-LAN can be a useful feature, but if you are not making use of it then it is simply draining extra power waiting for a magic packet while in suspend. You can adapt the **Wake-on-LAN#udev** rule to disable the feature for all ethernet interfaces. To enable powersaving with **iw** on all wireless interfaces:

```
/etc/udev/rules.d/81-wifi-powersave.rules
```

```
ACTION=="add", SUBSYSTEM=="net", KERNEL=="wl*", RUN+="/usr/bin/iw dev $name set power_save on"
```

The name of the configuration file is important. With the use of **persistent device names** in systemd, the above network rule, named lexicographically **after** `80-net-setup-link.rules`, is applied after the device is renamed with a persistent name e.g. `wlan0` renamed `wlp3s0`. Be aware that the `RUN` command is executed after all rules have been processed and must anyway use the persistent name, available in `$name` for the matched device.

3.7.1 Intel wireless cards (iwlwifi)

Additional power saving functions of Intel wireless cards with `iwlwifi` driver can be enabled by passing the correct parameters to the kernel module. Making them persistent can be achieved by adding the lines below to the `/etc/modprobe.d/iwlwifi.conf` file:

```
options iwlwifi power_save=1
```

This option will probably increase your median latency:

```
options iwlwifi uapsd_disable=0
```

On kernels < 5.4 you can use this option, but it will probably decrease your maximum throughput:

```
options iwlwifi d0i3_disable=0
```

Depending on your wireless card one of these two options will apply.

```
options iwlvm power_scheme=3
```

```
options iwldvm force_cam=0
```

You can check which one is relevant by checking which of these modules is running using

```
# lsmod | grep '^iwlmvm'
```

Keep in mind that these power saving options are experimental and can cause an unstable system.

3.8 Bus power management

3.8.1 Active State Power Management

From [Wikipedia](#):

Active-state power management (ASPM) is a power management mechanism for PCI Express devices to garner power savings while otherwise in a fully active state. Predominantly, this is achieved through active-state link power management; i.e., the PCI Express serial link is powered down when there is no traffic across it. It is normally used on laptops and other mobile Internet devices to extend battery life.

At boot, the BIOS enables or disables ASPM based on hardware support. To check for support:

```
# lspci -vv | grep 'ASPM.*abled;'
```

Fetch available ASPM policies and the current system default using the following:

```
$ cat /sys/module/pcie_aspm/parameters/policy  
  
[default] performance powersave powersupersave
```

ASPM might be disabled for the following reasons [\[3\]](#):

1. The BIOS disabled it for some reason (for conflicts?).
2. PCIE requires ASPM but L0s are optional (so L0s might be disabled and only L1 enabled).
3. The BIOS might not have been programmed for it or may be buggy.

If you believe that your hardware has support for ASPM despite the above, it can be force-enabled for the kernel to handle with the `pcie_aspm=force` [kernel parameter](#).

Warning:

- Force-enabling ASPM on an unsupported system may lead to increased power consumption. Furthermore, it may cause system freezes or kernel panics, so make sure you have a way to undo the option if it is unsuitable.
- Forcing ASPM takes place in the kernel, and therefore it may still remain disabled in hardware and not work. To check whether this is the case, run `dmesg | grep ASPM` as root. Consult the Wiki article specific to your hardware for more information if possible.

As long as ASPM is supported and enabled, it is possible to select a desired policy for the current session. For example, switch to `powersupersave` for the current session by doing the following:

```
# echo powersupersave > /sys/module/pcie_aspm/parameters/policy
```

To configure a specific ASPM state to enable upon system boot (using `powersupersave` as an example), add `pcie_aspm.policy=powersupersave` as a [kernel parameter](#).

3.8.2 PCI Runtime Power Management

```
/etc/udev/rules.d/pci_pm.rules
```

```
SUBSYSTEM=="pci", ATTR{power/control}="auto"
```

The rule above powers all unused devices down, but some devices will not wake up again. To allow runtime power management only for devices that are known to work, use simple matching against vendor and device IDs (use `lspci -nn` to get these values):

```
/etc/udev/rules.d/pci_pm.rules
```

```
# whitelist for pci autosuspend
SUBSYSTEM=="pci", ATTR{vendor}=="0x1234", ATTR{device}=="0x1234", ATTR{power/control}="auto"
```

Alternatively, to blacklist devices that are not working with PCI runtime power management and enable it for all other devices:

```
/etc/udev/rules.d/pci_pm.rules
```

```
# blacklist for pci runtime power management
SUBSYSTEM=="pci", ATTR{vendor}=="0x1234", ATTR{device}=="0x1234", ATTR{power/control}="on", GOTO="pci_pm_end"

SUBSYSTEM=="pci", ATTR{power/control}="auto"
LABEL="pci_pm_end"
```

3.8.3 USB autosuspend

The Linux kernel can automatically suspend USB devices when they are not in use. This can sometimes save quite a bit of power, however some USB devices are not compatible with USB power saving and start to misbehave (common for USB mice/keyboards). [udev](#) rules based on whitelist or blacklist filtering can help to mitigate the problem.

The most simple and likely useless example is enabling autosuspend for all USB devices:

```
/etc/udev/rules.d/50-usb_power_save.rules
```

```
ACTION=="add", SUBSYSTEM=="usb", TEST=="power/control", ATTR{power/control}="auto"
```

To allow autosuspend only for devices that are known to work, use simple matching against vendor and product IDs (use `lsusb` to get these values):

```
/etc/udev/rules.d/50-usb_power_save.rules
```

```
# whitelist for usb autosuspend
ACTION=="add", SUBSYSTEM=="usb", TEST=="power/control", ATTR{idVendor}=="05c6", ATTR{idProduct}=="9205", ATTR{power/control}="auto"
```

Alternatively, to blacklist devices that are not working with USB autosuspend and enable it for all other devices:

```
/etc/udev/rules.d/50-usb_power_save.rules
```

```
# blacklist for usb autosuspend
ACTION=="add", SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", ATTR{idProduct}=="9205", GOTO
="power_usb_rules_end"

ACTION=="add", SUBSYSTEM=="usb", TEST=="power/control", ATTR{power/control}="auto"
LABEL="power_usb_rules_end"
```

The default autosuspend idle delay time is controlled by the `autosuspend` parameter of the `usbcore` built-in [kernel module](#). To set the delay to 5 seconds instead of the default 2 seconds, add the following [kernel parameter](#) for your boot loader.

```
usbcore.autosuspend=5
```

Similarly to `power/control`, the delay time can be fine-tuned per device by setting the `power/autosuspend` attribute. This means, alternatively, autosuspend can be disabled by setting `power/autosuspend` to -1 (i.e., never autosuspend):

```
/etc/udev/rules.d/50-usb_power_save.rules
```

```
ACTION=="add", SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", ATTR{idProduct}=="9205", ATTR
{power/autosuspend}="-1"
```

See the [Linux kernel documentation](#) for more information on USB power management.

3.8.4 SATA Active Link Power Management



The factual accuracy of this article or section is disputed.

Reason: The warning at the start of the section makes it seem like all settings can lead to data loss, while the table implies it's only the lowest setting that may cause it. This should be explicitly clarified. (Discuss in [Talk:Power management](#))

Warning: SATA Active Link Power Management can lead to data loss on some devices. Do not enable this setting unless you have frequent backups.

Since Linux 4.15 there is a [setting](#) called `med_power_with_dipm` that matches the behaviour of Windows IRST driver settings and should not cause data loss with recent SSDs or HDDs. The power saving can be significant, ranging [from 1.0 to 1.5 Watts \(when idle\)](#). It has become the default setting for Intel based laptops in Linux 4.16 [\[4\]](#).

The current setting can be read from or written to

`/sys/class/scsi_host/host*/link_power_management_policy` as follows:

```
$ grep . /sys/class/scsi_host/host*/link_power_management_policy
$ echo "med_power_with_dipm" >/sys/class/scsi_host/hostN/link_power_management_policy
```

Available ALPM settings

Setting	Description	Power saving
max_performance	current default	None
medium_power	-	~1.0 Watts
med_power_with_dipm	recommended setting	~1.5 Watts
min_power	WARNING: possible data loss	~1.5 Watts

You can configure `link_power_management_policy` settings persistently by adding an [udev](#) rules file, for example:

```
/etc/udev/rules.d/hd_power_save.rules
```

```
ACTION=="add", SUBSYSTEM=="scsi_host", KERNEL=="host*", ATTR{link_power_management_policy}="med_power_with_dipm"
```

Note:

- This adds latency when accessing a drive that has been idle, so it is one of the few settings that may be worth toggling based on whether you are on AC power.
- Not all combinations of SATA host controllers and storage devices work well with the default `med_power_with_dipm` setting. For example, as of linux-6.8.1, Intel 7 Series Chipset controllers will experience timeouts and link degradation when accessing Crucial M550 SSDs (with firmware MU02). In such cases, using the `medium_power` setting can fix the problem.

3.9 Hard disk drive

See [hdparm#Power management configuration](#) for drive parameters that can be set.

Power saving is not effective when too many programs are frequently writing to the disk. Tracking all programs, and how and when they write to disk is the way to limit disk usage. Use [iotop](#) to see which programs use the disk frequently. See [Improving performance#Storage devices](#) for other tips.

Small adjustments such as setting the [noatime](#) option can also help. If enough RAM is available, consider disabling or limiting [swappiness](#) as it has the possibility to limit a good number of disk writes.

For Seagate drives with [PowerChoice](#) technology, tricks setting APM via [hdparm](#) will not work due to the *EPC* (Extended Power Conditions) feature. Rather than setting APM, you can [install openseachest](#)^{AUR} and fully disable EPC like so (replace *X* with actual drive letter):

```
# openSeaChest_PowerControl --scan
# openSeaChest_PowerControl -d /dev/sdX -i
# openSeaChest_PowerControl -d /dev/sdX --showEPCSettings
# openSeaChest_PowerControl -d /dev/sdX --EPCfeature disable
# openSeaChest_PowerControl -d /dev/sdX --showEPCSettings
```

Last invocation will give the following summary:

```
=====
```

```

===
openSeaChest_PowerControl - openSeaChest drive utilities - NVMe Enabled
Copyright (c) 2014-2023 Seagate Technology LLC and/or its Affiliates, All Rights Reserved
openSeaChest_PowerControl Version: 3.3.1-4_1_1 X86_64
Build Date: Jul  4 2023
Today: Tue Jul  4 17:49:36 2023          User: root
=====
===

```

```
/dev/sdX - ST1000NM0008-2F2100 - ZFA19JG2 - SN02 - ATA
```

===EPC Settings===

```

* = timer is enabled
C column = Changeable
S column = Savable
All times are in 100 milliseconds

```

Name	Current	Timer	Default	Timer	Saved	Timer	Recovery	Time	C	S
Idle A	0		*10		*10		1		Y	Y
Idle B	0		*1200		*1200		3		Y	Y
Idle C	0		6000		6000		16		Y	Y
Standby Z	0		9000		9000		46		Y	Y

Zeroes in the first column confirm that parking and spindown were disabled successfully

4 Tools and scripts



This article or section needs language, wiki syntax or style improvements. See [Help:Style](#) for reference.

Reason: Merged from [Power saving](#), needs reorganization to fit into this page. (Discuss in [Talk:Power management](#))

4.1 Using a script and an udev rule



This article or section is a candidate for merging with [Laptop#Power management](#).

Notes: Might be a better fit for the laptop-specific page. (Discuss in [Talk:Power management](#))

Since systemd users can suspend and hibernate through `systemctl suspend` or `systemctl hibernate` and handle acpi events with `/etc/systemd/logind.conf`, it might be interesting to remove *pm-utils* and **acpid**.

There is just one thing systemd cannot do (as of systemd-204): power management depending on whether the system is running on AC or battery. To fill this gap, you can create a single **udev** rule that runs a script when the AC adapter is plugged and unplugged:

```
/etc/udev/rules.d/powersave.rules
```

```

SUBSYSTEM=="power_supply", ATTR{online}=="0", RUN+="/path/to/your/script true"
SUBSYSTEM=="power_supply", ATTR{online}=="1", RUN+="/path/to/your/script false"

```

Note: You can use the same script that *pm-powersave* uses. You just have to make it executable and place it somewhere else (for example `/usr/local/bin/`).

Examples of powersave scripts:

- [ftw](#), package: [ftw-git](#)^{AUR}
- [powersave](#)
- [throttlectl](#), from [throttlectl](#)^{AUR}

The above udev rule should work as expected, but if your power settings are not updated after a suspend or hibernate cycle, you should add a script in `/usr/lib/systemd/system-sleep/` with the following contents:

```
/usr/lib/systemd/system-sleep/00powersave

#!/bin/sh

case $1 in
    pre) /path/to/your/script false ;;
    post)
        if cat /sys/class/power_supply/AC0/online | grep 0 > /dev/null 2>&1
        then
            /path/to/your/script true
        else
            /path/to/your/script false
        fi
    ;;
esac
exit 0
```

Do not forget to make it executable!

Note: Be aware that AC0 may be different for your laptop, change it if that is the case.

4.2 Print power settings



This article or section is a candidate for merging with [#Power saving](#).

Notes: Might be good as an introduction? Since most of the output of the script can be used by the following sections. (Discuss in [Talk:Power management](#))

This script prints power settings and a variety of other properties for USB and PCI devices. Note that root permissions are needed to see all settings.

```
#!/bin/bash

for i in $(find /sys/devices -name "bMaxPower")
do
    busdir=${i%/*}
    busnum=$(<$busdir/busnum)
    devnum=$(<$busdir/devnum)
    title=$(lsusb -s $busnum:$devnum)

    printf "\n\n+++ %s\n  -%s\n" "$title" "$busdir"
```

```

for ff in $(find $busdir/power -type f ! -empty 2>/dev/null)
do
    v=$(cat $ff 2>/dev/null|tr -d "\n")
    [[ ${#v} -gt 0 ]] && echo -e " ${ff##*/}=$v";
    v=;
done | sort -g;

done;

printf "\n\n\n+++ %s\n" "Kernel Modules"
for mod in $(lspci -k | sed -n '/in use:/s,^.*: ,,p' | sort -u)
do
    echo "+ $mod";
    systool -v -m $mod 2> /dev/null | sed -n "/Parameters:/,/^$/p";
done

```

5 Allow users to shutdown



This article or section needs language, wiki syntax or style improvements. See [Help:Style](#) for reference.

Reason: Merged from [Allow users to shutdown](#), needs reorganization to fit into this page. (Discuss in [Talk:Power management](#))

5.1 Button and lid events

The suspend, poweroff and hibernate button presses and lid close events are handled by *logind* as described in [#ACPI events](#).

5.2 Using systemd-logind

If you are using [polkit](#), users with non-remote session can issue power-related commands as long as [the session is not broken](#).

To check if your session is active

```
$ logindctl show-session $XDG_SESSION_ID --property=Active
```

The user can then use *systemctl* commands in the command line, or add them to menus:

```
$ systemctl poweroff
$ systemctl reboot
```

Other commands can be used as well, including `systemctl suspend` and `systemctl hibernate`. See the *System Commands* section in [systemctl\(1\)](#).

5.3 Using sudo

Install [sudo](#), and give the user [sudo privileges](#). The user will then be able to use the `sudo systemctl` commands (e.g. `sudo systemctl poweroff`, `sudo systemctl reboot`, `sudo systemctl suspend` and `sudo systemctl hibernate`). See the *System Commands* section in [systemctl\(1\)](#)

5.3.1 Users without sudo privileges

If users should only be allowed to use shutdown commands, but not have other sudo privileges, then, as root, add the following to the end of `/etc/sudoers` using the `visudo` command. Substitute *user* for your username and *hostname* for the machine's hostname.

```
user hostname =NOPASSWD: /usr/bin/systemctl poweroff,/usr/bin/systemctl halt,/usr/bin/systemctl reboot
```

Now your user can shutdown with `sudo systemctl poweroff`, and reboot with `sudo systemctl reboot`. Users wishing to power down a system can also use `sudo systemctl halt`. Use the `NOPASSWD:` tag only if you do not want to be prompted for your password.

6 See also

- [ThinkWiki:How to reduce power consumption](#)
- [How to get longer battery life on Linux](#)

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